

MATHEUS HOSTERT

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ACADEMIC POSITIONS

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| University of Iowa
Assistant Professor. | Aug. 2025 - present |
| Harvard University
Neutrino Theory Network Fellow. Department of Energy Fellowship. | Sept. 2023 - Aug. 2025 |
| University of Minnesota & Perimeter Institute
Joint four-year postdoctoral researcher appointment. | Sept. 2019 - Sept. 2023 |

EDUCATION

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|---|-----------------------|
| Ph.D. in Theoretical Physics – Durham University, United Kingdom
Dissertation: Hidden Physics at the Neutrino Frontier: Tridents, Dark Forces, and Hidden Particles.
Supervisor: Prof. Silvia Pascoli.
Dissertation Committee: Profs. Joachim Kopp and David Cerd  o. | Oct. 2015 - Sep. 2019 |
| Bachelor’s degree in Physics – Federal University of Santa Catarina, Brazil
With a year abroad at Durham University (Sept. 2013 - Sept. 2014) and honors in advanced mathematics. | Mar. 2011 - Jun. 2015 |

FELLOWSHIPS & AWARDS

- Old Gold Fellowship** (Apr. 2026): developmental program for tenure-track assistant and associate professors at the University of Iowa.
- Neutrino Theory Network fellowship** (Jan. 2023): fully funded postdoctoral position at Harvard University.
- Science without Borders Ph.D. scholarship** (Sept. 2015): excellence-based Brazilian scholarship for a full Ph.D. abroad.
- Science without Borders Undergraduate scholarship** (Sept. 2013): excellence-based Brazilian scholarship for one year of undergraduate studies abroad.

TEACHING

- **Courses Taught:**
 - Graduate-level Quantum Mechanics I, Fall 2026 (in progress).
 - Graduate-level Quantum Mechanics II, Spring 2026.
 - Graduate-level special topics course on Particle Physics Beyond-the-Standard Model, Fall 2025.
- **Other teaching:**
 - **International Neutrino Summer School 2026** (Summer 2026): lectures on “Beyond the Standard Model with Neutrinos”.
 - **Fermilab Neutrino University**: lecture on “New Physics with Neutrinos” as part of the Fermilab lecture series for summer interns.
 - **ICTP-SAIFR School on Dark Matter and Neutrino Detection, 2024**: three lectures on neutrinos beyond the Standard Model at ICTP-SAIFR in S  o Paulo.
 - **Perimeter ISSYP 2022 and 2023**: led high-school students in a one-week project about neutrinos.

SUPERVISION

- **Post-doctoral scholars:**

- Dr. Stephan Meighen-Berger (Fall 2025 – present), neutrinos and dark matter.
- Dr. Aparajitha Karthikeyan (Fall 2026 – present), Neutrino Theory Network Postdoctoral Fellow, neutrinos and dark matter.

- **Ph.D. students:**

- Ju-Yeol (Joel) Choi, working on accelerator neutrinos and future colliders.
- Esmenaz Isik, working on high-energy astrophysics and neutrino telescopes.
- Sean Chisolm, working on ultralight dark matter and neutrinos.

- **Undergraduate students:**

- Yatharth Sirohi, Fall 2025, Spring 2026, and Charles A. Wert Summer Research Grant 2026.
- Landen Freeman, Fall 2025, Spring 2026, and Charles A. Wert Summer Research Grant 2026.
- Megan Murphy, Fall 2025, Spring 2026, and Independent Creative Research by Undergraduates (ICRU) Fellowship 2026.

- **Postdoc mentoring in Ph.D. projects:**

- Jaime Hoefken (University of Bologna, 2022 to 2023): two publications.
- Daniele Massaro (University of Bologna, 2021 to 2023): four publications.
- Asli Abdullahi (Durham University, 2020 to 2021): two publications.
- Nicholas Kamp (MIT, 2021 to 2022): two publications.
- Nicolò Foppiani (Harvard University, 2020 to 2022): two publications.

- **Postdoc mentoring in undergraduate projects:**

- Isaac Cheng (University of Waterloo, 2023): primary advisor in PSI Start Summer project at the Perimeter Institute working on the cosmic neutrino background asymmetry.
- Nathan Gershengorn (Harvard University, 2024): research on the effect of matter potentials on the cosmic neutrino background asymmetry.
- Luc Bojorquez-Lopez (Harvard University, 2024): research on the beam-induced neutrino rate at muon collider detectors, resulting in a publication.
- Connor Sponsler (Harvard University, 2024): research on new matter resonances in neutrino propagation through the atmosphere, resulting in a publication.

RESEARCH THRUSTS & SELECTED PUBLICATIONS

Citation metrics:

- **h-index** of 19 (publications with 10 authors or less) / 44 (all publications)
- **Total citations** 1,223 (publications with 10 authors or less) / 9,622 (all publications)

The following is a selected list of publications for which I was one of the primary contributors. Author lists are displayed alphabetically, as is the standard in particle physics. A complete list can be found at inspirehep.net/authors/1621061.

★ marks papers led by a student or postdoc in my U. of Iowa research group.

☆ marks papers where I was the primary mentor.

Neutrinos at Short Baselines: Understanding the anomalous data from short-baseline neutrino experiments.

- **First Search for Dark Sector e^+e^- Explanations of the MiniBooNE Anomaly at MicroBooNE**, MicroBooNE collaboration, Matheus Hostert, Phys.Rev.Lett. **136** (2026) 12 121804, 7 citations.
- **Towards a Robust Exclusion of the Sterile-Neutrino Explanation of Short-Baseline Anomalies**, Ohana Benavides Rodrigues, Matheus Hostert, Kevin J. Kelly, Bryce Littlejohn, Pedro A. N. Machado, Ibrahim Safa, Tao Zhou, Phys.Rev.Lett. **135** (2025) 081801, arXiv:2503.13594, 5 citations.
- **MicroBooNE and the ν_e Interpretation of the MiniBooNE Low-Energy Excess**, Carlos A. Argüelles, Ivan Esteban, Matheus Hostert, Kevin J. Kelly, Joachim Kopp, Pedro A. N. Machado, Ivan Martinez-Soler, Yuber F. Perez-Gonzalez, Phys.Rev.Lett. **128** (2022) 241802, arXiv:2111.10359, 105 citations.
— *Editor's Suggestion* and featured in *Viewpoint: Neutrino Mystery Endures*.
- **Testing New Physics Explanations of the MiniBooNE Anomaly at Neutrino Scattering Experiments**, Carlos A. Argüelles, Matheus Hostert, Yu-Dai Tsai, Phys.Rev.Lett. **123** (2019) 261801, arXiv:1812.08768, 89 citations.

Neutrinos at Muon Colliders: Novel applications and sources of neutrinos at a multi-TeV muon collider.

- ★ **The Forward Neutrino Flux and its Secondaries at a 10 TeV Muon Collider**, Ju-Yeol Choi, Matheus Hostert, Peiran Li, Zhen Liu, under review at Phys.Rev.D, arXiv:2608.02718.
- ☆ **The Neutrino Slice at Muon Colliders**, Luc Bojorquez-Lopez, Matheus Hostert, Carlos A. Argüelles, Zhen Liu, Phys.Rev.Lett. **135** (2025) 091803, arXiv:2412.14115, 8 citations.

Spallation Sources: Novel applications of stopped-meson and stopped-muon sources for new physics.

- ☆ **Long-Lived Particles from Meson and Muon Decays at Rest at Spallation Sources**, Matheus Hostert, Salvador Urrea, under review at Rev. Mod. Phys., arXiv:2509.14085.
- **Probing Long-Lived Particle Production in Muon Decays at the SNS with a Highly Capable Hydrocarbon Detector**, PROSPECT collaboration, Matheus Hostert, Salvador Urrea, arXiv:2606.19299.

Astrophysical Neutrinos: Multi-messenger probes of neutrinos and dark sectors at low and high energies.

- ★ **New Multi-messenger Probe of Dark Matter-Nucleon Interactions from Ultra-High-Energy Cosmic Ray Acceleration**, Stephan A. Meighen-Berger, P. S. Bhupal Dev, Matheus Hostert, Phys.Rev.D **113** (2026) 103050, arXiv:2512.18093, 7 citations.
- **Neutrino-Antineutrino Conversion from Ultralight Vector Dark Matter**, Asher Berlin, Rodolfo Capdevilla, Ting Cheng, Matheus Hostert, Pedro A. N. Machado, under review at Phys.Rev.Lett., arXiv:2605.06777.
- **Standard Candles for Supernova Neutrino Detection at DUNE**, Ting Cheng, Matheus Hostert, Pedro A. N. Machado, Nityasa Mishra, Adrian Thompson, under review at Phys.Rev.Lett., arXiv:2606.18337.
- ☆ **Neutron Portal to Ultra-High-Energy Neutrinos**, Gustavo F. S. Alves, Matheus Hostert, Maxim Pospelov, Phys.Rev.D **113** (2026) 075035, arXiv:2503.14419, 20 citations.
- **Astrophysical Flux of Dark Particles as a Solution to the KM3NeT and IceCube Tension over KM3-230213A**, Yasaman Farzan, Matheus Hostert, JHEP **10** (2025) 208, arXiv:2505.22711, 17 citations.

Origin of Neutrino Masses: Novel approaches to understand and test low-scale neutrino mass models.

- **Dark Neutrinos and a Three Portal Connection to the Standard Model**, Peter Ballett, Matheus Hostert, Silvia Pascoli, Phys.Rev.D **101** (2020) 115025, arXiv:1903.07589, 111 citations.
- **Effective Portals to Heavy Neutral Leptons**, Enrique Fernández-Martínez, Manuel González-López, Josu Hernández-García, Matheus Hostert, Jacobo López-Pavón, JHEP **09** (2023) 001, arXiv:2304.06772, 107 citations.
- **Neutrino Masses from a Dark Neutrino Sector below the Electroweak Scale**, Peter Ballett, Matheus Hostert, Silvia Pascoli, Phys.Rev.D **99** (2019) 091701, arXiv:1903.07590, 59 citations.

Flavor and Lepton Precision: Light new physics in meson and charged-lepton decays.

- **Novel Multilepton Signatures of Dark Sectors in Light Meson Decays**, Matheus Hostert, Maxim Pospelov, Phys.Rev.D **105** (2022) 015017, arXiv:2012.02142, 48 citations.
- **Long-Lived Axion-Like Particles from Tau Decays**, Yohei Ema, Patrick J. Fox, Matheus Hostert, Tony Menzo, Jure Zupan, Phys.Rev.D **112** (2025) 115028, arXiv:2507.15271, 16 citations.
- **Muon-Induced Baryon Number Violation**, Patrick J. Fox, Matheus Hostert, Tony Menzo, Maxim Pospelov, Jure Zupan, Phys.Rev.D **110** (2024) 075015, arXiv:2407.03450, 11 citations.
— *Editor's Suggestion*.
- **New Physics in Multi-Electron Muon Decays**, Matheus Hostert, Tony Menzo, Maxim Pospelov, Jure Zupan, JHEP **10** (2023) 006, arXiv:2306.15631, 14 citations.

SELECTED TALKS

A full list of talks including plenary talks at international conferences, colloquia, invited conference and workshop talks, and invited seminars is available at www.mhostert.com/talks.

Convener Opening Talk, NuFact Workshops, Tsung-Dao Lee Institute, September 2026

Plenary Talk, APS Division of Particles and Fields (DPF) 2026, USA, July 2026

Plenary Talk, Neutrino 2026, USA, June 2026

Opening Theory Talk, New Physics Opportunities at Neutrino Experiments (NPN 2026), Ibaraki, Japan, May 2026

Plenary Talk, Phenomenology Symposium 2026, USA, May 2026

Invited Talk, APS Global Summit 2026 (Muon Collider Session), USA, April 2026

Invited Talk, APS Global Summit 2026 (Neutrino Session), USA, April 2026

Invited Talk, NPC Mini-Workshop on Short-Baseline Neutrino Anomalies, Fermilab, USA, January 2026

Invited Talk, Neutrino Physics Center Seminar Series, Fermilab, USA, January 2026

Colloquium, Columbia University, USA, December 2025

Invited Talk, Violation of Fundamental Symmetries with B mesons, Fermilab, USA, December 2025

Keynote Talk, Research Day at the Center for Neutrino Physics at Virginia Tech, USA, December 2025

Colloquium, Idaho State University, USA, October 2025

Convener Opening Talk, NuFact Workshop, University of Liverpool, September 2025.

Plenary Talk, 2nd US Muon Collider Meeting, USA, August 2025

Plenary Talk, Interconnection between Particle Physics and Cosmology (PPC2025), USA, June 2025

Invited Talk, NPN Cincinnati 2025, USA, June 2025

Invited Talk, CIPANP 2025, USA, June 2025

Invited Talk, Magnificent CEvNS 2025, Brazil (virtual), June 2025

Invited Talk, Mitchell Conference at Texas A&M, USA, May 2025

Invited Talk, Aspen Winter Conference, USA, March 2025

Convener Opening and Closing Talks, NuFact Workshop, Argonne National Laboratory, September 2024

Invited Talk, Muon Collider Meeting by the Lake, USA, September 2024

Invited Talk, Mitchell Conference at Texas A&M, USA, May 2024

Invited Talk, 2nd Short-Baseline Experiment-Theory Workshop, USA, April 2024

Colloquium, University of Iowa, USA, January 2024

Invited Talk, Muons in Minneapolis workshop, USA, December 2023

Invited Talk, American Physical Society April meeting, Minneapolis, USA, April 2023

SCIENCE OUTREACH

- **Media coverage:** Research and quotes featured in Quanta Magazine, The New York Times, and Wired.
- **IceCube Masterclass 2024:** main local organizer at Harvard for the IceCube Masterclass 2024.
- **Perimeter ISSYP 2022 and 2023:** tutor of the International Summer School for Young Physicists at Perimeter.

- **KITP Teacher’s Conference 2022**: speaker at the KITP teacher’s conference.
- Outreach during Ph.D.: **IceCube comics** translated all the Rosie & Gibbs comics about IceCube into Brazilian Portuguese, **Royal Society Summer Exhibition 2017** and **2018** event organizer for the “Modeling the Invisible” exhibition and the “Ghosts in the Universe” exhibition, **Celebrate Science 2018** volunteer in regional outreach event for schools in County Durham, **Orkney Science Festival 2018** volunteer in the International Orkney Science Festival, visiting schools in remote islands of the Orkney archipelago in the north of Scotland, **Pint of Science 2017** event manager for a local outreach event in County Durham.

COMMUNITY ENGAGEMENT

- **Violation of Fundamental Symmetries with B mesons**: co-convener for the Dark Sector sessions.
- **INSS 2025**: member of the local organizing committee for International Neutrino Summer School (INSS) 2025.
- **NuFact Workshop**: convener for “Neutrinos Beyond PMNS” working group under a three-year term (2024 - 2026).
- **Snowmass 2021**: Editor for the “Neutrino Frontier” white paper on sterile neutrinos and the “Rare Processes and Precision Measurements” white paper on new physics in kaon and hyperon factories. Made substantial contributions to over seven white papers and led a letter of intent on non-minimal neutrino mass models.
- **CERN FIP PBC**: member of the Feebly Interacting Particle (FIP) working group, part of the Physics Beyond Colliders (PBC) effort at CERN.
- **Weak Interactions and Neutrinos 2021**: one of primary event organizers for the 2021 edition in Minnesota, US.
- **IceDUNE workshop 2021**: convener of the Beyond the Standard Model physics session aimed at increasing the cross-talk between IceCube and DUNE collaborators.
- **Journal refereeing**: refereed for Physical Review Letters, Physical Review D, Journal of High Energy Physics, European Physics Journal C, and the Astrophysical Journal.

EXPERIMENTAL COLLABORATIONS

- Participated in a Memorandum of Understanding to collaborate with **MicroBooNE** on a search for dark sectors.
- Member of the **Deep Underground Neutrino Experiment (DUNE)** from 2015 to 2023 and of **IceCube** since 2023.

COMMUNITY PAPERS

A selected list of community papers to which I contributed.

1. **From oversimplified to overlooked: the case for exploring Rich Dark Sectors**, Invited contribution to the Nuclear Physics B Special Issue on “Clarifying common misconceptions in high energy physics and cosmology”, 2025, arXiv:2505.05663 [hep-ph], citations: 4.
2. **The present and future status of heavy neutral leptons**, Asli M. Abdullahi et al, J.Phys.G 50 (2023) 2 020501, arXiv:2203.08039 [hep-ph], citations: **348**.
3. **Feebly-interacting particles: FIPs 2020 workshop report**, Prateek Agrawal et al, Eur.Phys.J.C 81 (2021) 11 1015, 2021, arXiv:2102.12143, citations: **388**.
4. **Neutrino Non-Standard Interactions: A Status Report**, P. S. Bhupal Dev et al, SciPost Phys.Proc. 2 (2019) 001, 2019, arXiv:1907.00991, citations: **265**.
5. **White Paper on Light Sterile Neutrino Searches and Related Phenomenology**, M. A. Acero et al, preprint, 2022, arXiv:2203.07323, citations: **154**.
6. **New physics searches at kaon and hyperon factories**, Evgueni Goudzovski, et al, Rept.Prog.Phys. 86 (2023) 1 016201, arXiv:2201.07805 [hep-ph], citations: **177**.